Fiscal Year:	FY 2012	**Task Last Updated:**	FY 10/25/2012
PI Name:	Narayanan, Vinod Ph.D.		
Project Title:	Boiling Heat Transfer Mechanism During Thermally Actuated Pumping by Asymmetric Surface Structures		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Corvallis	**State:**	OR
Zip Code:	97331	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
Start Date:	06/01/2009	**End Date:**	09/30/2013
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
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Flight Program:			
Flight Assignment:	NOTE: Extended to 9/30/2013, per NSSC information (Ed., 12/18/2012) NOTE: Extension to 9/30/2013 is Pending, per NSSC and PI (Ed., 11/1/2012) NOTE: End date is 9/30/2012 per NSSC information (Ed., 2/28/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Bhavnani, Sushil (Co-PI: Auburn University)		
Grant/Contract No.:	NNX09AJ98G (OSU) ; NNX09AL63G (Auburn)		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Proposed is a collaborative research study between investigators at Oregon State University (OSU) and Auburn University (AU). A novel thermally actuated pumping mechanism during boiling heat transfer is identified in the context of passive means of enhancing heat transfer rate. The effect of an asymmetric force on the boiling process is the focus of the study. Such asymmetry is created by the use of a surface with repeated ratchet structures and pyramidal reentrant cavities located on one of its sides. The premise of the proposal is that surface asymmetry results in a force asymmetry during the growth and departure of the vapor bubbles causing local fluid motion in the streamwise direction (parallel to the surface). Forces that cause this fluid motion will be quantified at the microscale using minimally intrusive imaging methods. The hypothesis is that with several ratchets, this local motion can be translated to a net fluid pumping. The project has the following major tasks: 1) Fabricate ratchet –cavity structures surfaces using silicon microfabrication methods, 2) Perform proof-of-principle experiment, using high-speed flow visualization, confirming the bulk movement of the fluid by the proposed mechanism, 3) Measure, by analysis of High-Speed (HS) flow visualization movies, the time-varying bubble diameters and frequencies, 4) Determine, by temporally- and spatially- resolved temperature and heat flux measurements, heat transfer rate from the surface to the fluid in the small array ratchet experiments, and 5) Perform global measurements in the large array experiments. Program Significance: Passive methods of heat transfer enhancement are attractive for space and terrestrial applications since they use no external power and need little maintenance. The proposed phase-change enhancement technique can be applied for space thermal management systems and for enhancing heat transfer rate from adversely oriented surfaces. In microgravity pool boiling, it has been shown that the presence of a large primary vapor bubble near the heater surface causes premature dryout. Axial motion of this bubble by force asymmetry of the proposed concept could not only prevent local dryout but also enhance heat transfer. The proposed imaging methods present a detailed estimation of flow and temperature field in boiling flows that are of use to modellers. The proposed experiments and analyses will highlight forces of importance by varying orientations, subcooling and gravity levels, and use of fluids with contrasting properties. Broader Significance: Two graduate students will gain experience in the field of importance to NASA. Results from the project will be integrated in the form of a module into an existing Electronics Thermal Management live-internet multi-campus course at AU and will incorporate OSU as a new partner. Undergraduate students from both AU and OSU will be involved in the DC-9 flight experiments along with graduate students. The Oregon NASA Space Grant Consortium will help provide travel support for the DC-9 experiments as well as provide an avenue for dissemination of results to the community.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Program Significance: Passive methods of heat transfer enhancement are attractive for space and terrestrial applications since they use no external power and need little maintenance. The proposed phase-change enhancement technique can be applied for space thermal management systems and for enhancing heat transfer rate from adversely oriented surfaces. In microgravity pool boiling, it has been shown that the presence of a large primary vapor bubble near the heater surface causes premature dryout. Axial motion of this bubble by force asymmetry of the proposed concept could not only prevent local dryout but also enhance heat transfer. The proposed imaging methods present a detailed estimation of flow and temperature field in boiling flows that are of use to modellers. The proposed experiments and analyses will highlight forces of importance by varying orientations, subcooling and gravity levels, and use of fluids with contrasting properties.
Task Progress:	This is the third year of the grant. One journal paper is in print while two conference papers were presented. One graduate student at OSU graduated with an MS degree in June 2011. A second MS student has taken over the project on the OSU side. In Auburn one PhD student is working on the project and is anticipated to complete by December 2012. Both OSU and Auburn participated in a NASA fluid physics microgravity flight experiment the week of April 30th 2012. The hypothesis that bubbles emanating from cavities located on asymmetric microstructed surfaces causes a net axial fluid motion has been validated in 1g conditions. A preliminary model of bubble growth has been proposed to explain the fluid motion. Recent experiments under microgravity conditions and high g (1.8g) have also shown that large bubbles located near the microstructured surface are either rotated about a fixed position or moved preferentially along the microstructured surface.
Bibliography Type:	Description: (Last Updated: 02/03/2014)
Articles in Peer-reviewed Journals	Thiagarajan N, Kapsenberg F, Narayanan V, Bhavnani SH, Ellis CD. "Development of a heat sink with periodic asymmetric structures using gray-scale lithography and deep reactive ion etching." IEEE Electron Devices Letters, 2012 Jul;33(7):1054 - 6. http://dx.doi.org/10.1109/LED.2012.2197670 , Jul-2012
Dissertations and Theses	Kapsenberg F. "Lateral Fluid Motion in Nucleate Boiling Through Asymmetric Surface Structures." Masters Thesis, Oregon State University, June 2011. , Jun-2011
Papers from Meeting Proceedings	Thiagarajan N, Kapsenberg F, Narayanan V, Bhavnani S, Ellis C. "On The Lateral Motion of Bubbles Generated from Re-Entrant Cavities Located on Asymmetrically Structured Surfaces." ASME 2011 Pacific Rim Technical Conference and Exposition on Packaging and Integration of Electronic and Photonic Systems, InterPACK 2011, Portland, OR, July 6-8 2011. In: ASME 2011 Pacific Rim Technical Conference and Exhibition on Packaging and Integration of Electronic and Photonic Systems, MEMS and NEMS: Volume 2. Paper number Interpack 2011-52056, p. 123-131. http://dx.doi.org/10.1115/IPACK2011-52056 , Jul-2011

Papers from Meeting Proceedings	Kapsenberg F, Thiagarajan N, Narayanan V, Bhavnani SH. "Lateral Motion of Bubbles From Surfaces with Mini-ratchet Topography Modifications During Pool Boiling- Experiments and Preliminary Model." ITherm 2012, 13th IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems, San Diego, CA, May 30-June 1, 2012. ITherm 2012, 13th IEEE Intersociety Conference on Thermal and Thermomechanical Phenomena in Electronic Systems, San Diego, CA, May 30-June 1, 2012. Paper# 3062. , May-2012